

Research Idea: Overlap

ALMA Thesis Planner (automated pipeline)

Title: A shock-tracer census of the Antennae overlap region: what fraction of column-selected dense clumps show SiO/HNCO, and where does the shocked gas sit in velocity?

Data. ALMA 2013.1.01041.S alone (public since 2015-11-12; one member OUS ‘uid://A001/X12a/X4e’; Band 6 mosaic of the Antennae overlap region; $0.268''$ native beam ≈ 30 pc at the Antennae’s distance, 1.33 km/s channels, 0.975 mJy/beam per 10 km/s, continuum sensitivity 0.030 mJy/beam). Archive QA2 cubes first; CASA re-imaging only if the QA2 products prove unusable for the two faint shock lines. No new observations, no external catalogs, surveys, or papers.

Question. The proposal’s full aim — a pre-cluster-cloud (PCC) population with chemical formation timescales — is paper-length. This thesis takes its measurable core:

1. Among dense clumps above a fixed column-density cut, lying inside the region of the mosaic where the SiO limit reaches a fixed uniform depth, **what fraction show SiO(5–4) or HNCO(10_{0,10}–9_{0,9}) emission?** 2. For the (probably majority) non-detections, **where in velocity does the ensemble-stacked shock signal sit relative to the bulk gas?** SiO from a cloud–cloud collision need not appear at the CO centroid; a shifted or broad-winged stacked profile is itself a statement about how the dense gas is being assembled.

Pre-registered headline configuration (single, fixed before analysis). $^{13}\text{CO}(2-1)$ catalog, $0.5''$ uv-tapered cube. Every headline number is quoted from this one configuration. The native- $0.27''$ version and a C^{18}O -based cross-check appear once, in a single appendix robustness table, and nowhere else.

Work plan (one student, ~nine months).

1. *Week 1 — literature check.* Confirm no published SiO/HNCO census exists from this OUS. The census + velocity-resolved stack survives either way.

2. *Months 1–2 — cubes, hard handoff date.* Line IDs verified in Splatalogue. Products: SiO(5–4) ~ 216.06 GHz, HNCO ~ 218.62 , $\text{C}^{18}\text{O}(2-1) \sim 218.38$, $^{13}\text{CO}(2-1) \sim 219.22$, $\text{H}_2\text{CO}(3_{0,3}-2_{0,2})/\text{CH}_3\text{OH} \sim 217.1-217.3$ GHz, plus a 1.3 mm continuum image from the 231–235 GHz windows. Headline cubes are uv-tapered to $0.5''$ (≈ 55 pc, the PCC scale the proposal targeted); the expected brightness-sensitivity gain over native resolution is $\sim 2\times$, not more, since down-weighting long baselines raises per-beam noise, and the sample-depth cut in step 4 is built on the measured noise map, not on a predicted factor. Month 3 begins on whatever cubes exist on the handoff date.

3. *Month 3 — one clump catalog.* Dendrogram/CPROPS-style segmentation on the $^{13}\text{CO}(2-1) \ 0.5''$ cube — the only line reliably above the noise at clump scales. $\text{C}^{18}\text{O}(2-1)$ supplies velocity centroids and an optical-depth check for the clumps where it is detected; it is not a segmentation input. Each entry carries position, deconvolved size, velocity centroid, linewidth,

peak and integrated ^{13}CO intensity, and a uv-filtering diagnostic (aperture flux vs. surrounding negative-bowl depth). Expect a few tens of objects.

4. *Month 4 — a denominator defined by physics.* The census sample is clumps whose ^{13}CO -derived peak column exceeds a stated fixed cut **and** which lie inside the mosaic region where the SiO 3σ limit reaches a stated uniform depth, read off the per-pixel noise map. Everything else is excluded from the fraction and tabulated separately. This makes the fraction reproducible and, in principle, comparable to other datasets.

5. *Months 5–6 — measurement.* Aperture spectra of SiO and HNCN at every surviving position: an integrated intensity or a 3σ upper limit, each with the local noise attached.

6. *Months 7–9 — three results.* (a) **Column-selected detected fraction**, with binomial confidence intervals, always quoted in the same sentence as the column cut and the limiting depth. (b) **Velocity-resolved stack.** Clump spectra velocity-aligned to their C^{18}O (else ^{13}CO) centroids and co-added, with the stacking window wide enough to capture ± 30 km/s offsets. The headline figure is **stacked SiO (and HNCN) intensity as a function of velocity offset from the centroid**, with the stacked S/N shown against integration-window width. Ratios are reported as *intensity* ratios $I(\text{SiO})/I(^{13}\text{CO})$, $I(\text{HNCN})/I(^{13}\text{CO})$ — not abundances, since one transition constrains no excitation. Secondary split: linewidth quartiles, as binned stacks only. (c) **Column-matched selectivity test.** Does grain/shock chemistry track something other than column? Clumps are split into H_2CO -bright and H_2CO -faint halves **at fixed ^{13}CO integrated intensity** — i.e. matched in bins of $I(^{13}\text{CO})$, so the comparison is between clumps of the same column — and the stacked SiO/ ^{13}CO ratio compared between the matched samples. The discriminating quantity is stated in advance: an elevated **$I(\text{H}_2\text{CO})/I(^{13}\text{CO})$** ratio, not raw H_2CO brightness, is what counts as “grain chemistry”; a threshold in that ratio is fixed before the SiO stacks are examined. If the H_2CO -bright and H_2CO -faint samples turn out not to overlap in column, the test is reported as inconclusive rather than forced.

Cut deliberately, and stated as such: no second clump catalog, no survival analysis (the censoring level scales as SiO noise / ^{13}CO intensity, i.e. with the very covariate of interest, so a Kaplan–Meier ratio distribution would be biased in an unignorable direction), and no shock-peak/clump-centroid offset statistics (with the expected handful of detections, any such test is powerless; individual detections are presented as case studies with no statistical claim). Also out of scope: shock-model grids, chemical clocks, PCC formation timescales, non-LTE radiative transfer, and any global gas-mass budget, which single-configuration 12 m filtering makes indefensible. The 1.3 mm continuum is used only to flag clumps as possibly hosting embedded HII regions, since Band 6 alone cannot separate dust from free-free. Optional stretch, time permitting: LTE column densities for the brightest few SiO clumps across an assumed T_{ex} range, with the assumption’s effect displayed.

Deliverables. The 0.5'' cubes and moment maps, one machine-readable ^{13}CO clump catalog with filtering flags and SiO/HNCN intensities-or-limits, three figures (detected fraction with binomial errors; stacked SiO/HNCN versus velocity offset; column-matched H_2CO -bright vs. H_2CO -faint SiO/ ^{13}CO comparison), and a one-page appendix robustness table repeating the fraction for the native-resolution and C^{18}O variants.

Why it is tractable. Public, calibrated, modest-volume data; archive-first imaging bounds the schedule risk; standard CASA and Python tooling throughout; one pre-registered configuration means one headline number rather than four. The design produces a quotable result about a canonical merger in the most likely outcome — SiO faint across most of the overlap region —

because a measured fraction, a stacked limit resolved in velocity, and a column-matched chemistry comparison are all measurements, not detections.